

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials (other than your formula sheet).
- Show all of your work, and use correct notation. Full credit will not be given without sufficient supporting work.
 - If you use a theorem, rule, definition, etc., tell me that you're using it!
- Round all numeric answers to at least 2 decimal places.
- You are awesome, and I believe in you!

1. (14 pts.) Let Y be a continuous random variable with the following pdf:

$$f(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

- (a) (8 pts.) Find $F(y)$.

Case 1 ($y < 0$): $F(y) = P(Y \leq y) = \int_{-\infty}^y 0 dt = 0$

Case 2 ($0 \leq y \leq 1$): $F(y) = P(Y \leq y) = \int_{-\infty}^0 0 dt + \int_0^y 3t^2 dt = t^3 \Big|_0^y = y^3$

Case 3 ($y > 1$): $F(y) = P(Y \leq y) = \int_{-\infty}^0 0 dt + \int_0^1 3t^2 dt + \int_1^y 0 dt = t^3 \Big|_0^1 = 1$

$$F(y) = \begin{cases} 0, & y < 0 \\ y^3, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

- (b) (6 pts.) Calculate $P(0.4 \leq Y \leq 0.7)$.

$$P(0.4 \leq Y \leq 0.7) = F(0.7) - F(0.4) = (.7)^3 - (.4)^3 = \underline{\underline{.279}}$$

2. (8 pts.) A discrete random variable Y has the following cumulative distribution function:

$$F(y) = \begin{cases} 0, & y < 1 \\ 0.3, & 1 \leq y < 6 \\ 0.5, & 6 \leq y < 10 \\ 0.9, & 10 \leq y < 12 \\ 1, & y \geq 12 \end{cases}$$

Find the probability distribution, $p(y)$, of Y .

y	1	6	10	12
$p(y)$	0.3	0.2	0.4	0.1
	$\underbrace{\hspace{1cm}}$	$\underbrace{\hspace{1cm}}$	$\underbrace{\hspace{1cm}}$	$\underbrace{\hspace{1cm}}$
	.3-0	.5-.3	.9-.5	1-.9

3. (16 pts.) According to the *National Health and Nutrition Examination Survey*, published by the National Center for Health Statistics, the serum (noncellular portion of blood) total cholesterol level of U.S. females 20 years or older is normally distributed with a mean of 206 milligrams per deciliter (mg/dL) and standard deviation of 44.7 mg/dL.

- (a) (8 pts.) What proportion of U.S. females 20 years or older have cholesterol levels below 200?

Let X = cholesterol level.

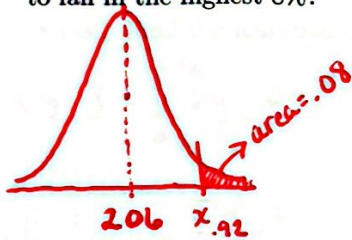
$$P(X < 200) = P\left(Z < \frac{200 - 206}{44.7}\right) = P(Z < -.1342)$$

$$= \text{normalcdf}(-1E99, -.1342, 0, 1)$$

$$\approx \underline{\underline{.4466}}$$

OR $\text{normalcdf}(-1E99, 200, 206, 44.7) \approx \underline{\underline{.4466}}$

- (b) (8 pts.) How high must the cholesterol level of a U.S. female 20 years or older be in order to fall in the highest 8%?



$$\text{Want } \bar{x}_{.92} = \text{invNorm}(.92, 206, 44.7)$$

$$\approx \underline{\underline{268.8067 \text{ mg/dL}}}$$

4. (8 pts.) Consider the random variable Y , which has a probability density function given by

$$f(y) = \begin{cases} ky^2 e^{-2y}, & y > 0 \\ 0, & \text{o.w.} \end{cases}$$

Find the value of k that makes $f(y)$ a valid probability density function. Hint: integrating is not the quickest approach here.

By inspection, $Y \sim \text{Gamma}(\alpha = 3, \beta = 1/2)$.

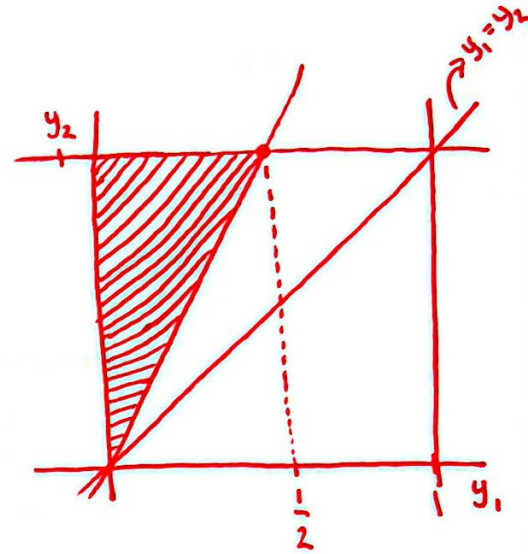
$$\text{Therefore, } k = \frac{1}{\beta^\alpha \Gamma(\alpha)} = \frac{1}{(\frac{1}{2})^3 \Gamma(3)} = \frac{1}{(\frac{1}{8})(2!)} = \underline{\underline{4}}.$$

5. (24 pts.) Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 3y_2, & 0 \leq y_1 \leq y_2 \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

(a) (8 pts.) Calculate $P(Y_2 \geq 2Y_1)$.

$$\begin{aligned} P(Y_2 \geq 2Y_1) &= \int_0^{1/2} \int_{2y_1}^1 3y_2 \, dy_2 \, dy_1 \\ &= \int_0^{1/2} \left[\frac{3}{2} y_2^2 \Big|_{y_2=2y_1}^{y_2=1} \right] dy_1 \\ &= \int_0^{1/2} \left(\frac{3}{2} - 6y_1^2 \right) dy_1 \\ &= \left[\frac{3}{2} y_1 - 2y_1^3 \right]_0^{1/2} \\ &= \frac{3}{4} - \frac{1}{4} = \boxed{\frac{1}{2}} \end{aligned}$$



(b) (8 pts.) Find the marginal density function of Y_2 .

$$f_2(y_2) = \int_0^{y_2} 3y_2 \, dy_1 = 3y_2 y_1 \Big|_{y_1=0}^{y_1=y_2} = 3y_2^2, \quad 0 \leq y_2 \leq 1$$

$$\Rightarrow f_2(y_2) = \begin{cases} 3y_2^2, & 0 \leq y_2 \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

(c) (8 pts.) Calculate $P(Y_1 \leq 1/8 \mid Y_2 = 3/4)$.

$$\begin{aligned} P(Y_1 \leq 1/8 \mid Y_2 = 3/4) &= \int_0^{1/8} \frac{f(t_1, 3/4)}{f_2(3/4)} \, dt_1 \\ &= \int_0^{1/8} \frac{(9/4)}{(27/16)} \, dt_1 \\ &= \frac{4}{3} \left[t_1 \Big|_0^{1/8} \right] \\ &= \frac{4}{3} \left[\frac{1}{8} - 0 \right] \\ &= \boxed{\frac{1}{6}} \end{aligned}$$

6. (6 pts.) Identify the distribution of the random variable with the following moment-generating function. Be specific—label and give values for the parameters of the distribution.

$$m(t) = e^{20t + 3t^2}$$

By inspection, this is the mgf of a Normal distribution.

$$\cdot \underline{\underline{\mu = 20}}$$

$$\cdot \frac{\sigma^2}{2} = 3 \Rightarrow \sigma^2 = 6 \Rightarrow \underline{\underline{\sigma = \sqrt{6}}}$$

7. (14 pts.) Suppose X_1 and X_2 are random variables with the following properties:

$$\begin{array}{lll} E(X_1) = -2 & V(X_1) = 4 & Cov(X_1, X_2) = 2 \\ E(X_2) = 1 & V(X_2) = 3 & \end{array}$$

- (a) (8 pts.) Let $Y = 2X_1 - 3X_2$ and $W = X_1 + X_2$. Calculate $Cov(Y, W)$.

$$\begin{aligned} Cov(Y, W) &= Cov(2X_1 - 3X_2, X_1 + X_2) \\ &= 2(1)Cov(X_1, X_1) + 2(1)Cov(X_1, X_2) - 3(1)Cov(X_2, X_1) - 3(1)Cov(X_2, X_2) \\ &= 2V(X_1) + 2Cov(X_1, X_2) - 3Cov(X_1, X_2) - 3V(X_2) \\ &= 2(4) + 2(2) - 3(2) - 3(3) \\ &= -3 \end{aligned}$$

- (b) (6 pts.) Are Y and W independent? Justify your answer.

No! If Y and W were independent, then $Cov(Y, W)$ would equal 0 (by Theorem at bottom of Handout 21). We just showed that $Cov(Y, W) = -3$.

Choose one of problems 8-10 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may select which one to grade.

Handout 21

8. (10 pts.) Prove the following result: $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2)$. You may use the following notation: $\mu_1 = E(Y_1)$ and $\mu_2 = E(Y_2)$.

$$\begin{aligned}
 \text{Cov}(Y_1, Y_2) &= E[(Y_1 - \mu_1)(Y_2 - \mu_2)] \quad [\text{by definition of covariance}] \\
 &= E[Y_1 Y_2 - Y_1 \mu_2 - \mu_1 Y_2 + \mu_1 \mu_2] \\
 &= E(Y_1 Y_2) - \mu_2 E(Y_1) - \mu_1 E(Y_2) + \mu_1 \mu_2 \\
 &= E(Y_1 Y_2) - E(Y_2)E(Y_1) - \cancel{E(Y_1)E(Y_2)} + \cancel{E(Y_1)E(Y_2)} \\
 &= E(Y_1 Y_2) - E(Y_1)E(Y_2). \quad \square
 \end{aligned}$$

Handout 15

9. (10 pts.) Suppose $X \sim \exp(\beta)$ and $a, b > 0$. Prove that $P(X > a + b \mid X > a) = P(X > b)$.

$$\begin{aligned}
 P(X > a + b \mid X > a) &= \frac{P(X > a + b \cap X > a)}{P(X > a)} \quad [\text{def. of conditional prob.}] \\
 &= \frac{P(X > a + b)}{P(X > a)} \\
 &= \frac{1 - P(X \leq a + b)}{1 - P(X \leq a)} \quad [\text{Complement Rule}] \\
 &= \frac{1 - F(a + b)}{1 - F(a)} \\
 &= \frac{1 - (1 - e^{-(a+b)/\beta})}{1 - (1 - e^{-a/\beta})} \\
 &= \frac{e^{-(a+b)/\beta}}{e^{-a/\beta}} \\
 &= e^{-b/\beta} \\
 &= P(X > b). \quad \square
 \end{aligned}$$

10. (10 pts.) Let Y_1 and Y_2 be independent, $g(Y_1)$ be a function of Y_1 , and $h(Y_2)$ be a function of Y_2 . Prove the following:

$$E[g(Y_1)h(Y_2)] = E[g(Y_1)] \cdot E[h(Y_2)].$$

Let $f(y_1, y_2)$ denote the joint density of Y_1 and Y_2 .

The product $g(Y_1)h(Y_2)$ is a function of Y_1 and Y_2 .

$$\text{Thus, } E[g(Y_1)h(Y_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(y_1)h(y_2)f(y_1, y_2)dy_2dy_1,$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(y_1)h(y_2)f_1(y_1)f_2(y_2)dy_2dy_1 \quad [\text{by independence}]$$

$$= \int_{-\infty}^{\infty} g(y_1)f_1(y_1) \underbrace{\left[\int_{-\infty}^{\infty} h(y_2)f_2(y_2)dy_2 \right]}_{E[h(Y_2)]} dy_1,$$

$$= E[h(Y_2)] \cdot \underbrace{\int_{-\infty}^{\infty} g(y_1)f_1(y_1)dy_1}_{E[g(Y_1)]},$$

$$= E[h(Y_2)] \cdot E[g(Y_1)]. \quad \square$$